7 Hedge Funds

- **Why do asset pricing researchers study hedge funds?**
  - **The sharpest test of "skill":** hedge funds are the least constrained, most highly paid managers. If anyone can beat the market, they should. But their returns are **option-like**, **illiquid** and **self-reported**, so **measuring alpha is the research problem**.
  - **A laboratory for risk factors:** dynamic, leveraged, long-short strategies carry risks that buy-and-hold indices miss; hedge fund returns forced the literature to build **new benchmarks**.
  - **A data problem in itself:** voluntary reporting creates biases, and illiquid holdings produce smoothed or misreported returns that can manufacture "alpha."
  - **Berk–Green in its purest form:** capacity-constrained strategies, observable flows and explicit fee contracts make hedge funds the natural place to watch **capital chase alpha and erode it**.

7.1 Institutional Background

Feature	Hedge funds
Legal form	Private partnerships exempt from the Investment Company Act; accredited investors only; large advisers file Form PF with the SEC since 2012
Size	About $5 trillion of net assets worldwide (Barth, Joenväärä, Kauppila, and Wermers, 2021); roughly 10,000 funds, concentrated in a few hundred large managers
Fees	"2 and 20": management fee on assets plus an **incentive fee** on profits above a **high-water mark** (and sometimes a hurdle rate); in practice 1.5 and 17–19
Liquidity terms	**Lock-ups** (often 1 year), redemption notice periods, redemption frequency (monthly, quarterly), gates, side pockets
Strategies	Long/short equity, event driven, relative value and convertible arbitrage, fixed income, global macro, managed futures (CTAs), multi-strategy, funds of funds
Tools	Short selling, leverage, derivatives, illiquid assets; dynamic, state-dependent trading
Data	**Voluntary** monthly net returns and AUM reported to vendors (TASS, HFR, BarclayHedge, CISDM, Eurekahedge, Preqin); **mandatory** quarterly 13F long equity holdings (managers >$100M)

7.2 The Key Research Questions

1. **What do hedge funds do, and what risks do they bear?** Dynamic, leveraged, long-short strategies with **option-like** payoffs.
2. **How should we measure hedge fund performance?** Which benchmark—and from returns or from holdings? Alpha is only as good as the benchmark.
3. **Can we trust the data?** Voluntary reporting (survivorship, backfill, self-selection); **smoothed** and **misreported** returns.
4. **Is there alpha, where does capital go, and does capital erode alpha?** Luck vs. skill; the Berk–Green view; what do investors chase?
5. **Do contracts and incentives matter?** Incentive fees, high-water marks, ownership, lock-ups: alignment or risk shifting?
6. **How do liquidity, funding and leverage shape returns and crises?** Liquidity premia; forced deleveraging (1998, 2007, 2008).
7. **Are hedge funds smart money?** Do their holdings and trades correct mispricing (anomalies, bubbles, activism) or ride it?

Questions 1–4 are covered in detail below; 5–7 only through the map on the next two slides.

7.2 A map of the hedge fund literature

Question	Key papers
Risk exposures and performance measurement	Sharpe (1992); Fung and Hsieh (1997)★; Fung and Hsieh (2001)★; Mitchell and Pulvino (2001); Agarwal and Naik (2004)★; Fung and Hsieh (2004)★; Bollen and Whaley (2009); Patton and Ramadorai (2013); Bali, Brown, and Caglayan (2014); holdings-based: Griffin and Xu (2009); Grinblatt, Jostova, Petrasek, and Philipov (2020)★
Data biases	Brown, Goetzmann, and Ibbotson (1999); Ackermann, McEnally, and Ravenscraft (1999); Fung and Hsieh (2000); Liang (2000); Malkiel and Saha (2005); Aiken, Clifford, and Ellis (2013); Agarwal, Fos, and Jiang (2013); Jorion and Schwarz (2019); Joenväärä, Kauppila, Kosowski, and Tolonen (2021); Barth, Joenväärä, Kauppila, and Wermers (2021)
Return quality: smoothing and misreporting	Asness, Krail, and Liew (2001); Lo (2002); Getmansky, Lo, and Makarov (2004)★; Bollen and Pool (2009)★; Cassar and Gerakos (2011); Agarwal, Daniel, and Naik (2011); Bollen and Pool (2012)★; Patton, Ramadorai, and Streatfield (2015)
Alpha, persistence, capital flows	Kosowski, Naik, and Teo (2007); Fung, Hsieh, Naik, and Ramadorai (2008)★; Jagannathan, Malakhov, and Novikov (2010); Dichev and Yu (2011); Getmansky (2012); Agarwal, Green, and Ren (2018)★; Bollen, Joenväärä, and Kauppila (2021)

Papers marked ★ are covered in detail below; the topics on the next slide are not covered in detail.

7.2 A map of the hedge fund literature (continued)

Question	Key papers
Contracts and incentives	Goetzmann, Ingersoll, and Ross (2003); Agarwal, Daniel, and Naik (2009); Panageas and Westerfield (2009); Aragon and Nanda (2012); Lim, Sensoy, and Weisbach (2016); Ben-David, Birru, and Rossi (2023)
Liquidity, funding, leverage, crises	Mitchell, Pedersen, and Pulvino (2007); Khandani and Lo (2007); Aragon (2007); Sadka (2010); Teo (2011); Ang, Gorovyy, and Inwegen (2011); Boyson, Stahel, and Stulz (2010); Aragon and Strahan (2012); Ben-David, Franzoni, and Moussawi (2012); Cao, Chen, Liang, and Lo (2013); Jylhä, Rinne, and Suominen (2014)
Smart money: holdings and trades	Brunnermeier and Nagel (2004); Brav, Jiang, Partnoy, and Thomas (2008); Griffin and Xu (2009); Agarwal, Jiang, Tang, and Yang (2013); Kokkonen and Suominen (2015); Cao, Liang, Lo, and Petrasek (2018)

Surveys: Agarwal, Mullally, and Naik (2015); Getmansky, Lee, and Lo (2015).

7.3 Tools: Measuring Hedge Fund Performance

- **Same question as for mutual funds, harder data.** We still ask whether the fund beats a **passive alternative**; but for hedge funds
  - **only net returns are observed** (monthly, self-reported); holdings are available only for long U.S. equity positions (13F) and only for some funds, so **returns-based** methods dominate;
  - strategies are **dynamic and nonlinear**: betas change within the month, and payoffs resemble options, so a constant-beta linear model on buy-and-hold indices misses the risk;
  - histories are **short** and returns are **non-normal and smoothed**, so alphas are imprecise and standard inference fails.
- **The answer the literature converged on:** benchmark hedge funds with **asset-based style factors**—returns on *tradable* strategies (including option strategies) that mimic what the funds do—and then correct for data biases before claiming alpha.
- **The sequence of papers** that built the benchmark is on the next slide; it is also our first set of papers.

7.3 A sequence of papers: from style regressions to the seven-factor model

Paper	Method	Key finding	Problem it leaves → next paper
Sharpe (1992)	**Returns**: regress fund returns on asset-class indices (weights ≥ 0, summing to one)	Mutual fund returns are mostly explained by a few asset classes	Assumes buy-and-hold, long-only, constant weights: none holds for hedge funds
Fung and Hsieh (1997)	**Returns**: (i) Sharpe's regression; (ii) **principal component analysis** of fund returns: five **style factors**, each a portfolio of the funds most correlated with one component	Asset classes explain little ($R^2 < 25\%$ for half the funds). Five style factors: Systems/Trend Following, Systems/Opportunistic, Global/Macro, Value, Distressed	Style factors are **portfolios of hedge funds**, not traded assets: they show *which* funds move together, not *what* market risk a style bears
Fung and Hsieh (2001)	**Returns**: trend following modeled as **lookback straddles** on 26 futures markets	Five straddle portfolios explain 48% of trend-follower returns; long volatility	One style only; what about the equity-oriented strategies?
Agarwal and Naik (2004)	**Returns**: add S&P 500 **call and put** returns to buy-and-hold factors; stepwise selection	Arbitrage strategies load negatively on puts: they **write crash insurance**	Many candidate factors; need a parsimonious, common benchmark
Fung and Hsieh (2004)	**Returns**: **seven asset-based factors**: equity (2), bond (2), trend-following straddles (3)	R^2 of 0.55–0.80 for fund-of-funds indices; alpha insignificant in subperiods	Constant betas (Bollen and Whaley, 2009; Patton and Ramadorai, 2013); omitted risks (liquidity: Sadka, 2010) look like alpha

7.3 Returns-based factor models for hedge funds: what they can and cannot do

	Returns-based (Fung–Hsieh, option factors)	**Holdings-based (13F)**
Data	Monthly net returns, voluntarily reported	Quarterly long U.S. equity positions, mandatory for managers >$100M
Pros	Covers all strategies and asset classes; net of fees; option factors capture nonlinear payoffs; straddle factors are **tradable**, so alpha has a clear meaning	No self-reporting biases, no smoothing; shows *what* funds hold and trade (bubbles, anomalies, activism); characteristic benchmarks as for mutual funds
Cons	Constant betas vs. dynamic trading (Bollen and Whaley, 2009; Patton and Ramadorai, 2013); omitted factors (liquidity, volatility) look like alpha; short, smoothed, biased return series; many candidate factors ⇒ data mining	Long equity only: no shorts, derivatives, leverage or bonds; quarterly snapshots; only a subset of funds; before fees and trading costs
Used in	Fung–Hsieh (1997, 2001, 2004), Agarwal–Naik, Kosowski–Naik–Teo, Fung–Hsieh–Naik–Ramadorai	Brunnermeier–Nagel, Griffin–Xu, Brav–Jiang–Partnoy–Thomas, Grinblatt–Jostova–Petrasek–Philipov

Bottom line: the hedge fund literature is mostly **returns-based by necessity**; holdings papers measure style and skill for the long-equity part of the industry.

7.4 Three Sets of Papers and How They Build on Each Other

Set	Paper	Question	Key method / result used later
Risk exposures and performance measures	Fung and Hsieh (1997)	Are hedge funds buy-and-hold?	Style regressions fail; five style factors from principal components
	Fung and Hsieh (2001)	How to model a dynamic strategy?	Trend followers as **lookback straddles**: option-based factors
	Agarwal and Naik (2004)	Which nonlinear risks?	Index **call and put** returns as factors; arbitrage strategies write puts
	Fung and Hsieh (2004)	A common benchmark?	The **seven-factor model**: the standard hedge fund benchmark
	Grinblatt, Jostova, Petrasek, and Philipov (2020)	Style and skill from holdings?	GTW/GT measures on 13F holdings: contrarian hedge funds have skill
Return quality	Getmansky, Lo, and Makarov (2004)	Why serially correlated returns?	MA model of **smoothed returns**: unsmooth before measuring risk
	Bollen and Pool (2009) and Bollen and Pool (2012)	Are returns misreported?	**Kink at zero** in pooled returns; return-pattern fraud flags
Alpha and capital flows	Fung, Hsieh, Naik, and Ramadorai (2008)	Who has alpha; what do flows do to it?	Have-alpha vs. beta-only funds; inflows erode alpha (**Berk–Green**)
	Agarwal, Green, and Ren (2018)	Which alpha do investors chase?	Flow horse race (Barber, Huang, and Odean, 2016): CAPM alpha wins

7.5 Fung and Hsieh (1997): Hedge Funds Are Not Buy-and-Hold

- **Method 1—Sharpe's style regression on hedge funds:** regress each fund's monthly return on eight asset-class indices (equities, bonds, dollar, gold, cash).
- **Result:** it fails. More than half of mutual funds have $R^2 > 75\%$; **48% of hedge funds have** $R^2 < 25\%$, and 25% are *negatively* correlated with the asset classes.
- **Why—style has two dimensions: location** (which asset classes, the F's) and **trading strategy** (direction and leverage, the w's, anywhere between -10 and $+10$ and changed quickly). A manager who flips between long and short S&P futures each month has a coefficient near zero although all his risk is in U.S. equities.
- **Method 2—let the data define the styles: principal components** of fund returns; the portfolio of funds most correlated with each component is a **style factor**. Five style factors capture the common variation.

7.5 Fung-Hsieh (1997): Five styles, and the first sign of option-like payoffs

- **The five style factors:** **Systems/Trend Following**, **Systems/Opportunistic**, **Global/Macro**, **Value**, **Distressed**. Value and Distressed load on equities and high-yield bonds (*location* styles); the two Systems styles have no asset-class exposure (*trading-strategy* styles).
- **Nonlinearity:** trend followers earn their best returns in both the *worst* and the best equity months—a payoff that is not linear in the market and that a regression coefficient cannot summarize.
- **What later papers take from it:**
  - linear models on buy-and-hold indices understate hedge fund risk, so low R^2 and large alphas are **not** evidence of skill;
  - styles should be defined by *what funds do*, with **tradable** style factors added to the asset classes—the program carried out in Fung and Hsieh (2001) and Fung and Hsieh (2004);
  - option-like payoffs need option-like factors → Agarwal and Naik (2004).

7.6 Fung and Hsieh (2001): Trend Following as a Lookback Straddle

- **Goal:** model one dynamic style **from first principles** and build **tradable** factors for it. Trend followers are the natural case: their best months are the **worst and the best** equity months (Figure 1). Over an interval, let S, S' be the start and end prices, and $S_{\max}$, $S_{\min}$ the high and low. With one trade per interval:

Strategy	Best payoff	Equivalent option
Market timer (Merton, 1981)	$\lvert S' - S \rvert$	standard straddle
Trend follower (PTFS)	$S_{\max} - S_{\min}$	lookback straddle

- A **lookback straddle** buys at the low and sells at the high (Goldman, Sosin, and Gatto, 1979). It is replicated by **rolling two straddles**: roll one up when the price makes a new high and one down at a new low—just like "buying breakouts and selling breakdowns."
- **Construction:** lookback straddles on **26 futures markets**, grouped into five **primitive trend-following strategy (PTFS)** factors: stocks, bonds, interest rates, currencies, commodities.

7.6 Fung-Hsieh (2001): Results

Adjusted R^2 of trend-follower returns on alternative factors (Table 2)

Factors	Adjusted R^2
Eight standard asset classes (Fung and Hsieh, 1997)	1.0%
Commodity indices (GSCI, CRB)	< 0
Buy-and-hold in the same 26 markets (by group)	-3 to 8%
Five PTFS portfolios	**47.9%**

- **Key finding:** buy-and-hold factors explain nothing; lookback straddles explain about half. In the 1987 and 1990 equity crashes, trend followers **gained** 10–13%.
- **Takeaway, used by everything that follows:** trend followers have real systematic risk—they are **long volatility**—and it can be captured by **returns on tradable option strategies**. The PTFS become three of the seven factors of the standard benchmark.

7.7 Fung and Hsieh (2004): The Seven-Factor Model

- **Idea:** peer-group hedge fund indices inherit database biases; instead, benchmark hedge funds with **asset-based style (ABS) factors** built from market prices:

Strategy	Factors
Equity long–short	(1) S&P 500; (2) small minus large cap
Fixed income	(3) Δ10-year yield; (4) Δcredit spread
Trend following	(5)–(7) bond, currency, commodity PTFS

- **Key finding:** the seven factors explain 55–80% of a fund-of-funds index, and the remaining alpha is **insignificant** once exposures are allowed to shift after 1998 and 2000.
- **Why it matters:** the seven-factor model became the **standard hedge fund benchmark** (e.g., Fung, Hsieh, Naik, and Ramadorai, 2008; Kosowski, Naik, and Teo, 2007).

7.8 Agarwal and Naik (2004): Nonlinear Risks of Hedge Funds

- **Question:** are **option-like (nonlinear) payoffs** pervasive across equity-oriented hedge fund strategies—not only trend followers (Fung and Hsieh, 2001) and merger arbitrage (Mitchell and Pulvino, 2001)?
- **Motivating fact:** arbitrage (non-directional) strategies have high means and low volatility but strongly negative skewness and fat tails—the signature of selling insurance.
- **Approach:** add returns on **traded index options** to the factor model, so a nonlinear payoff becomes a linear loading on an option return.

7.8 Agarwal-Naik (2004): Methodology

- **Asymmetry test:** $R_t^i = \alpha^i + \beta_0^i RUS_t + \beta_1^i D_t + \gamma^i D_t \times RUS_t + \varepsilon_t^i$, with $D_t = 1$ if the market is below its median. Most strategies have about **zero beta in up markets and positive beta in down markets**.
- **Option-based factors:** Glosten and Jagannathan (1994) show nonlinear payoffs can be approximated with returns on index options:

$$R_p = \alpha + \beta_1 R_m + \sum_j \beta_{c,j} \max(R_m - k_j, 0) + \sum_j \beta_{p,j} \max(k_j - R_m, 0) + \varepsilon \quad (8)$$

- Factors: **ATM and OTM S&P 500 calls and puts** plus buy-and-hold factors (equity, SMB, HML, MOM, bonds, credit, FX, commodities), selected by **stepwise regression**:

$$R_t^i = c^i + \sum_{k=1}^{K} \lambda_k^i F_{k,t} + u_t^i \quad (9)$$

- **Out-of-sample:** portfolios built from in-sample loadings track the strategy indices, so the exposures are not an artifact.

7.8 Agarwal-Naik (2004): Hedge funds sell crash insurance

HFR strategy index (Table 4)	Option factor loading	Other significant factors	Adj. R^2
Event arbitrage	OTM put −0.92	SMB, HML	44%
Restructuring	OTM put −0.63	SMB, HML, high yield, EM	66%
Event driven	OTM put −0.94	Market, SMB, HML, EM	73%
Relative value arb.	OTM put −0.64	MOM (−), SMB, HML	52%
Convertible arb.	ATM put −0.27	lagged market, SMB, bonds	41%
Equity hedge	none	Market, SMB, HML (−)	73%
Short selling	OTM call −1.38	Market (−), SMB (−), HML	82%

- **Five of six non-directional strategies load negatively on put options**—their payoffs resemble **writing put options** (selling portfolio insurance): small gains in calm markets, large losses in crashes.
- **Used later:** index option returns as hedge fund risk factors; mean-variance analysis understates the tail risk of these strategies, and alphas measured against linear models overstate skill.

7.9 Grinblatt, Jostova, Petrasek, and Philipov (2020): Measuring Style and Skill from Holdings

- **Research questions:**
  1. Mutual funds are momentum investors (Carhart, 1997; Grinblatt, Titman, and Wermers, 1995). Do **hedge funds**—the most sophisticated and incentivized managers—also trade momentum, or are they **contrarian**?
  2. How is **style** related to **skill**? Is performance persistent?
  3. When hedge funds and mutual funds trade against each other, **who is on the right side**?
- **Why holdings?** The returns-based benchmarks above rely on self-reported returns. Mandatory **13F holdings** avoid self-reporting, smoothing, survivorship and backfill biases, and allow the **holdings-based measures of the mutual fund literature** (GTW 1995, GT 1993, DGTW 1997) to be applied to hedge funds and mutual funds side by side.

7.9 GJPP (2020): Measuring style and skill from holdings

- **Style:** the GTW (1995) lag-0 momentum measure for adviser i in quarter q:

$$L0M_{iq} = \sum_{j=1}^{N(q)} (w_{i,j,q} - w_{i,j,q-1})\, R_{j,q} \tag{10}$$

  $L0M > 0$: buys stocks that went up during the quarter (momentum); $L0M < 0$: contrarian.

- **Skill:** the Grinblatt-Titman (1993) measure using **next-quarter DGTW-adjusted returns**:

$$F1M^*_{iq} = \sum_{j} (w_{i,j,q} - w_{i,j,q-1})\, R^*_{j,q+1}, \qquad R^*_{j,q+1} = R_{j,q+1} - R^{DGTW}_{j,q+1} \tag{11}$$

  Double benchmarking (prior holdings *and* DGTW characteristics) $\Rightarrow$ performance is *not* just momentum.

- **Tests:** classify advisers by the sign of $L0M$; compare $F1M$ across groups; test persistence; compute alphas of stocks each group buys and sells.

7.9 GJPP (2020): Style—hedge funds are contrarian

(Table 2, Panel A)

L0M, % per quarter (t)	Mutual funds		Hedge funds	
	All	Buys	All	Buys
Time-series average	0.71 (14.88)	0.39 (5.94)	−0.34 (−3.62)	−0.37 (−2.43)
Proportion contrarian	0.33	0.40	0.65	0.67

- **Mutual funds are momentum investors:** they buy recent winners.
- **Hedge funds are contrarian**—two-thirds of hedge fund advisers—mostly by **buying losers**.
- **Style is persistent:** about four in five contrarian hedge funds (momentum mutual funds) keep their style.

7.9 GJPP (2020): Skill—only contrarian hedge funds have it

(Tables 3 and 4)

F1M, % per quarter (t)	Mutual funds	Hedge funds
All, size/BM-adjusted	0.13 (2.67)	0.27 (3.61)
All, DGTW-adjusted (incl. momentum)	0.02 (0.51)	0.25 (3.72)
Contrarian advisers, DGTW-adjusted	0.01 (0.31)	**0.34 (4.09)**
Momentum advisers, DGTW-adjusted	0.02 (0.42)	0.08 (0.86)

- **Mutual fund "skill" is momentum:** it disappears once the benchmark controls for momentum.
- **Contrarian hedge funds are the only managers with stock-picking skill** (about 1.4–2.4% per year before fees), and it **persists**.
- **Who is right?** The stocks mutual funds sell outperform—and contrarian hedge funds are the ones buying them.

7.10 Return Quality: Can We Trust the Data?

- **Every regression so far uses self-reported returns.** Funds report to databases **voluntarily**, mostly as marketing; each decision to start or stop reporting is **informative**, and biases the sample:
  - **Survivorship bias:** funds that stop reporting drop out of the "live" file; returns overstated by 2–3%/yr (Fung and Hsieh, 2000; Liang, 2000).
  - **Backfill bias:** a fund joins *after* a good incubation period and its history is backfilled: about 1.4%/yr; drop pre-**listing-date** returns (Jorion and Schwarz, 2019).
  - **Self-selection, both ways:** funds stop reporting *before* bad returns (Aiken, Clifford, and Ellis, 2013); the best funds never report—in Form PF data non-reporting funds **outperform** (Barth, Joenväärä, Kauppila, and Wermers, 2021).
- **Standard practice:** live + dead funds; drop pre-listing returns; merge databases. **Next:** smoothing and misreporting.

7.11 Getmansky, Lo, and Makarov (2004): Serial Correlation and Smoothed Returns

- **The puzzle:** monthly hedge fund returns are often **highly serially correlated** (autocorrelations of 30–50% for some funds), which time-varying expected returns or leverage cannot generate. **Thesis: illiquidity and smoothed marks**.
- **Method:** **observed** returns are a moving average of true returns $R_t = \mu + \beta\Lambda_t + \varepsilon_t$,

$$R_t^o = \theta_0 R_t + \theta_1 R_{t-1} + \cdots + \theta_k R_{t-k}, \qquad \theta_j \in [0,1], \qquad \sum_{j=0}^{k} \theta_j = 1 \tag{12}$$

  estimated by maximum likelihood as an MA(2) process; smoothing index $\xi = \sum_j \theta_j^2$.
- **Implications** (Proposition 1): the mean is unaffected, but $\text{Var}[R_t^o] = \xi\sigma^2 < \sigma^2$: **volatility and betas are understated**, the Sharpe ratio is overstated by $1/\sqrt{\xi}$, and beta is spread over lags, $\beta_m^o = \theta_m \beta$.

7.11 GLM (2004): Smoothing is concentrated in illiquid styles

Hedge fund category (MA(2) estimates; Table 8)	$\hat{\theta}_0$	$\hat{\theta}_1$	$\hat{\theta}_2$	$\hat{\xi}$
Fixed-income directional	0.76	0.15	0.08	0.67
Event-driven	0.81	0.15	0.04	0.74
Relative value	0.82	0.09	0.09	0.82
US equity hedge	0.95	0.06	−0.01	1.00
Managed futures	1.13	−0.05	−0.08	1.47

- **Key result:** smoothing is concentrated where assets are **illiquid**; fixed-income funds report only 76% of the current month's true return, and naive Sharpe ratios are overstated by 16–20% in illiquid categories.
- **Used later:** **unsmooth** returns (invert the MA) or add **lagged factors** before estimating betas, alphas and Sharpe ratios; $\hat{\theta}_0$ is a proxy for illiquidity. GLM cannot tell illiquidity from *deliberate* smoothing—the next papers can.

7.12 Bollen and Pool (2009): Do Managers Misreport Returns?

- **Question:** is deliberate **misreporting** widespread, not just isolated fraud? Incentives: flows reward the **number of positive months**, and managers of illiquid assets have **discretion** in valuation.
- **Method** (from the earnings-management literature): if small losses are reported as small gains, the **pooled distribution** of monthly returns has a **discontinuity at zero**. Fit a smooth kernel density $\hat{f}(t;h) = \frac{1}{Nh}\sum_i \phi\left(\frac{x_i - t}{h}\right)$ as the null, integrate it over each bin to get p, and compare with the actual count x:

$$z = \frac{x - Np}{\sqrt{Np(1-p)}} \sim N(0,1) \quad \text{under a smooth density} \qquad (13)$$

- **Controls:** the same test on mutual funds, CTAs and factor-fitted returns, which have no discretion to misreport.

7.12 Bollen-Pool (2009): Results—a kink at zero that vanishes near audits

z-statistics for the two bins next to zero ($|z| > 2.58$: significant at 1%; Table VI)

Sample	Bin below 0	Bin above 0
All hedge funds	−15.27	15.02
Returns fitted from 10 risk factors	−0.59	0.12
3 months ending in an audit	−1.32	0.33
Equity market neutral (liquid)	−0.09	3.62
Distressed securities (illiquid)	−5.00	3.78

- **Misreporting, not strategy or data biases:** the kink is absent for mutual funds, CTAs and factor-fitted returns, **vanishes near audits**, and is stronger in **illiquid** funds.
- **Overstatements reverse:** two-month returns show no kink, and returns after a small gain are *lower* than after a small loss—the opposite of GLM smoothing. About **10% of returns** appear distorted (Figure 5B).

7.13 Bollen and Pool (2012): Suspicious Return Patterns Predict Fraud

- **Question:** can **return patterns alone** flag funds later charged with fraud (SEC actions or investor lawsuits for misreporting)?
- **Method—performance flags,** each with a trigger set by simulation:
  - **return-characteristic flags:** low correlation with style factors; **serial correlation** (GLM); the **kink at zero** (BP 2009);
  - **data-pattern flags:** too few negative returns; too many zero, repeated or too few unique returns; non-uniform last digits.
- **Results:** fraud funds trigger almost every flag more often than other funds; nearly half of the misreporting funds score in the top decile of a composite flag. False positives are common: a **screen**, not a verdict.
- **For measurement:** run these **diagnostics** before trusting an alpha; biases, smoothing and misreporting all push measured performance **up**.

7.14 Alpha and Capital Flows: The Berk–Green View of Hedge Funds

- **With a benchmark and clean data in hand, the mutual fund questions return:** is there alpha, where does capital go, and does capital erode alpha?
- **Why hedge funds are the better test of Berk and Green (2004):** capacity constraints are real (arbitrage opportunities are finite), flows are observable, and the fee contract makes the division of rents explicit.
- **Luck vs. skill, in one line:** with thousands of funds and short, non-normal histories, the top alphas can be luck. Kosowski, Naik, and Teo (2007) show with a bootstrap that funds ranked on $t(\alpha)$ are not merely lucky and that alpha persists—**weakest among funds with high inflows**, the capacity effect we study next.
- **Two papers:** Fung, Hsieh, Naik, and Ramadorai (2008)—who has alpha and what inflows do to it; Agarwal, Green, and Ren (2018)—which alpha investors chase when they send the money. (Who keeps the alpha: incentive fees take about half of gross profits, Ben-David, Birru, and Rossi, 2023.)

7.15 Fung, Hsieh, Naik, and Ramadorai (2008): Performance, Risk, and Capital Formation

- **Questions:** (1) Which funds have alpha, and does alpha persist? (2) Where does investor capital go? (3) Does the arrival of capital **reduce** alpha, as decreasing returns to scale (Berk and Green, 2004) predict?
- **Why funds of funds?** Their returns are net of *all* fees, reflect what an investor could actually earn, and suffer less from reporting biases.
- **Benchmark:** the Fung and Hsieh (2004) seven-factor model, with **structural breaks** in exposures at October 1998 (LTCM) and April 2000 (end of the tech bubble).

7.15 FHNR (2008): Have-alpha vs. beta-only funds

- **Classification:** every year, estimate the seven-factor model on the prior **two years** of each fund; a fund is **have-alpha** if its alpha is significantly positive by the Kosowski, Timmermann, Wermers, and White (2006) bootstrap, and **beta-only** otherwise.
- **How many have alpha?** About **one in five** funds of funds.
- **Does it persist?** Modestly: a have-alpha fund stays have-alpha twice as often as a beta-only fund becomes one (28% vs. 14%), and have-alpha funds are far less likely to be liquidated.
- **Does it last?** The alpha of have-alpha funds **fell sharply after 2000**, exactly when the industry's assets exploded.

7.15 FHNR (2008): Capital flows and the erosion of alpha

Flows and subsequent alpha by fund type (Tables 5–7)

	Have-alpha funds	Beta-only funds
Average annual net inflow	29.7%	8.2%
Flow response to lagged returns	0.20 (insignificant)	0.33 (significant)
Stays have-alpha, below-median inflows	34%	—
Stays have-alpha, above-median inflows	22%	—
Next $t(\alpha)$, below vs. above-median inflows	1.84 vs. 1.47 ($p = 0.01$)	no difference

- **Capital finds the alpha producers:** have-alpha funds receive far larger and **steadier** inflows; it is the beta-only funds whose flows **chase raw past returns**.
- **And the capital erodes it:** among have-alpha funds, those with above-median inflows are less likely to keep their alpha and have lower future $t(\alpha)$. No such effect for beta-only funds, whose alpha was never there.

7.15 FHNR (2008): Interpretation and what came after

- **Berk–Green in hedge funds:** skilled managers exist, investors identify them and send money, and the money **competes the alpha away**—through capacity limits in arbitrage strategies rather than price impact alone. The post-2000 decline in alpha coincides with the industry's growth from a niche to a $1–2 trillion business.
- **Versus Kosowski, Naik, and Teo (2007):** KNT find persistence at the fund level with precise (Bayesian) alphas; FHNR find persistence at the FoF level that inflows weaken. Both: alpha is real, scarce, and capacity-constrained.
- **Used later:** the have-alpha/beta-only classification with bootstrapped seven-factor alphas; the "end of an era" of hedge fund alpha (Bollen, Joenväärä, and Kauppila, 2021); which model investors use when they send the money—next slide.

7.16 Agarwal, Green, and Ren (2018): What Drives Hedge Fund Flows?

- **Question:** which risk model do hedge fund investors use when they allocate capital—and do they distinguish alpha from returns earned by **traditional** risk exposures (market, size, value, bonds) and **exotic** ones available mainly through hedge funds (momentum, options, trend following)?
- **Method—the mutual fund tools applied to hedge funds** (Barber, Huang, and Odean, 2016; Berk and van Binsbergen, 2016):
  - the **sign test**: how often does next year's flow have the sign of this year's alpha under each model?
  - the **pairwise horse race** across six models (CAPM, FF3, Carhart, Agarwal–Naik, Fung–Hsieh 7, a 12-factor model);
  - the **return decomposition**: regress flows on alpha, the traditional-beta return and the exotic-beta return.

7.16 Agarwal-Green-Ren (2018): CAPM alpha wins again—and investors chase exotic betas

Sign test (Table 3) and return decomposition (Table 7): annual flow on last year's return components

Sign agreement, flow vs. alpha		Flow coefficients by model			
Measure	%	Model	Alpha	Traditional β	Exotic β
CAPM alpha	61.1	Carhart 4-factor	11.8	5.5	6.2
Raw return	60.8	Agarwal–Naik	10.8	6.3	7.2
Carhart alpha	59.8	Fung–Hsieh 7	11.2	5.5	9.5
FF3 alpha	59.0	12-factor	10.4	7.1	8.3

- **CAPM alpha explains flows best** (sign test; it wins every pairwise horse race, Table 5), as for mutual funds—even though hedge fund investors are institutions and pay performance fees.
- **Investors chase all three components**, and the **exotic-beta** return at least as much as the traditional one (significantly more for the Fung–Hsieh model). Neither component's return **persists** (Table 8): chasing recent exotic-risk returns is a poor strategy.

7.17 What Have We Learned?

- **Dynamic traders need dynamic benchmarks.** Style regressions fail for hedge funds (Fung and Hsieh, 1997); trend followers are **lookback straddles** (Fung and Hsieh, 2001); arbitrageurs **write puts** (Agarwal and Naik, 2004); the seven asset-based factors (Fung and Hsieh, 2004) turn much of the "alpha" of the 1990s into beta. From holdings, contrarian hedge funds have real stock-picking skill (Grinblatt, Jostova, Petrasek, and Philipov, 2020).
- **Fix the data before measuring alpha.** Reporting is a choice (survivorship, backfill, self-selection); returns are **smoothed** (Getmansky, Lo, and Makarov, 2004) and sometimes **misreported** (Bollen and Pool, 2009; Bollen and Pool, 2012). All three push measured performance up.
- **Alpha is real, scarce, and capacity-constrained.** Capital finds the have-alpha funds and erodes their alpha (Fung, Hsieh, Naik, and Ramadorai, 2008); investors judge funds by **CAPM alpha** and chase exotic betas (Agarwal, Green, and Ren, 2018)—Berk and Green, with the fee contract doing part of the work.

Reading List for Hedge Funds

Vikas Agarwal, T. Clifton Green, and Honglin Ren. "Alpha or Beta in the Eye of the Beholder: What Drives Hedge Fund Flows?" In: *Journal of Financial Economics* 127.3 (2018), pp. 417–434.

Vikas Agarwal and Narayan Y. Naik. "Risks and Portfolio Decisions Involving Hedge Funds". In: *The Review of Financial Studies* 17.1 (Jan. 2004), pp. 63–98.

Nicolas P. B. Bollen and Veronika K. Pool. "Do Hedge Fund Managers Misreport Returns? Evidence from the Pooled Distribution". In: *Journal of Finance* 64.5 (2009), pp. 2257–2288.

Nicolas P. B. Bollen and Veronika K. Pool. "Suspicious Patterns in Hedge Fund Returns and the Risk of Fraud". In: *Review of Financial Studies* 25.9 (2012), pp. 2673–2702.

William Fung and David A. Hsieh. "Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds". In: *Review of Financial Studies* 10.2 (Summer 1997), pp. 275–302.

William Fung and David A. Hsieh. "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers". In: *Review of Financial Studies* 14.2 (Summer 2001), pp. 313–341.

William Fung and David A. Hsieh. "Hedge Fund Benchmarks: A Risk-Based Approach". In: *Financial Analysts Journal* 60.5 (September–October 2004), pp. 65–80.

William Fung, David A. Hsieh, Narayan Y. Naik, and Tarun Ramadorai. "Hedge Funds: Performance, Risk, and Capital Formation". In: *Journal of Finance* 63.4 (2008), pp. 1777–1803.

Mila Getmansky, Andrew W. Lo, and Igor Makarov. "An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns". In: *Journal of Financial Economics* 74.3 (Dec. 2004), pp. 529–609.

Mark Grinblatt, Gergana Jostova, Lubomir Petrasek, and Alexander Philipov. "Style and Skill: Hedge Funds, Mutual Funds, and Momentum". In: *Management Science* 66.12 (Dec. 2020), pp. 5505–5531.

References for Hedge Funds

1 Carl Ackermann, Richard McEnally, and David Ravenscraft. "The Performance of Hedge Funds: Risk, Return, and Incentives". In: *Journal of Finance* 54.3 (June 1999), pp. 833–874.

2 Vikas Agarwal, Naveen D. Daniel, and Narayan Y. Naik. "Role of Managerial Incentives and Discretion in Hedge Fund Performance". In: *Journal of Finance* 64.5 (Oct. 2009), pp. 2221–2256.

3 Vikas Agarwal, Naveen D. Daniel, and Narayan Y. Naik. "Do Hedge Funds Manage Their Reported Returns?" In: *Review of Financial Studies* 24.10 (2011), pp. 3281–3320.

4 Vikas Agarwal, Vyacheslav Fos, and Wei Jiang. "Inferring Reporting-Related Biases in Hedge Fund Databases from Hedge Fund Equity Holdings". In: *Management Science* 59.6 (2013), pp. 1271–1289.

5 Vikas Agarwal, T. Clifton Green, and Honglin Ren. "Alpha or Beta in the Eye of the Beholder: What Drives Hedge Fund Flows?" In: *Journal of Financial Economics* 127.3 (2018), pp. 417–434.

6 Vikas Agarwal, Wei Jiang, Yuehua Tang, and Baozhong Yang. "Uncovering Hedge Fund Skill from the Portfolio Holdings They Hide". In: *Journal of Finance* 68.2 (Apr. 2013), pp. 739–783.

7 Vikas Agarwal, Kevin A. Mullally, and Narayan Y. Naik. "The Economics and Finance of Hedge Funds: A Review of the Academic Literature". In: *Foundations and Trends in Finance* 10.1 (2015), pp. 1–111.

8 Vikas Agarwal and Narayan Y. Naik. "Risks and Portfolio Decisions Involving Hedge Funds". In: *The Review of Financial Studies* 17.1 (Jan. 2004), pp. 63–98.

9 Adam L. Aiken, Christopher P. Clifford, and Jesse Ellis. "Out of the Dark: Hedge Fund Reporting Biases and Commercial Databases". In: *Review of Financial Studies* 26.1 (2013), pp. 208–243.

10 Andrew Ang, Sergiy Gorovyy, and Gregory B. van Inwegen. "Hedge Fund Leverage". In: *Journal of Financial Economics* 102.1 (2011), pp. 102–126.

11 George O. Aragon. "Share Restrictions and Asset Pricing: Evidence from the Hedge Fund Industry". In: *Journal of Financial Economics* 83.1 (2007), pp. 33–58.

12 George O. Aragon and Vikram Nanda. "Tournament Behavior in Hedge Funds: High-Water Marks, Fund Liquidation, and Managerial Stake". In: *Review of Financial Studies* 25.3 (2012), pp. 937–974.

13 George O. Aragon and Philip E. Strahan. "Hedge Funds as Liquidity Providers: Evidence from the Lehman Bankruptcy". In: *Journal of Financial Economics* 103.3 (2012), pp. 570–587.

14 Clifford Asness, Robert Krail, and John Liew. "Do Hedge Funds Hedge?" In: *Journal of Portfolio Management* 28.1 (2001), pp. 6–19.

15 Turan G. Bali, Stephen J. Brown, and Mustafa O. Caglayan. "Macroeconomic Risk and Hedge Fund Returns". In: *Journal of Financial Economics* 114.1 (2014), pp. 1–19.

16 Brad M. Barber, Xing Huang, and Terrance Odean. "Which Factors Matter to Investors? Evidence from Mutual Fund Flows". In: *Review of Financial Studies* 29.10 (2016), pp. 2600–2642.

17 Daniel Barth, Juha Joenväärä, Mikko Kauppila, and Russ Wermers. "The Hedge Fund Industry Is Bigger (and Has Performed Better) Than You Think". Working paper 20-01, Office of Financial Research. 2021.

18 Itzhak Ben-David, Justin Birru, and Andrea Rossi. "The Performance of Hedge Fund Performance Fees". In: *Journal of Finance* 78.3 (2023), pp. 1549–1588.

19 Itzhak Ben-David, Francesco Franzoni, and Rabih Moussawi. "Hedge Fund Stock Trading in the Financial Crisis of 2007-2009". In: *Review of Financial Studies* 25.1 (Jan. 2012), pp. 1–54.

20 Jonathan B. Berk and Richard C. Green. "Mutual Fund Flows and Performance in Rational Markets". In: *Journal of Political Economy* 112.6 (Dec. 2004), pp. 1269–1295.

21 Jonathan B. Berk and Jules H. van Binsbergen. "Assessing Asset Pricing Models Using Revealed Preference". In: *Journal of Financial Economics* 119.1 (2016), pp. 1–23.

22 Nicolas P. B. Bollen, Juha Joenväärä, and Mikko Kauppila. "Hedge Fund Performance: End of an Era?" In: *Financial Analysts Journal* 77.3 (2021), pp. 109–132.

23 Nicolas P. B. Bollen and Veronika K. Pool. "Do Hedge Fund Managers Misreport Returns? Evidence from the Pooled Distribution". In: *Journal of Finance* 64.5 (2009), pp. 2257–2288.

24 Nicolas P. B. Bollen and Veronika K. Pool. "Suspicious Patterns in Hedge Fund Returns and the Risk of Fraud". In: *Review of Financial Studies* 25.9 (2012), pp. 2673–2702.

25 Nicolas P. B. Bollen and Robert E. Whaley. "Hedge Fund Risk Dynamics: Implications for Performance Appraisal". In: *Journal of Finance* 64.2 (2009), pp. 985–1035.

26 Nicole M. Boyson, Christof W. Stahel, and René M. Stulz. "Hedge Fund Contagion and Liquidity Shocks". In: *Journal of Finance* 65.5 (2010), pp. 1789–1816.

27 Alon Brav, Wei Jiang, Frank Partnoy, and Randall Thomas. "Hedge Fund Activism, Corporate Governance, and Firm Performance". In: *Journal of Finance* 63.4 (2008), pp. 1729–1775.

28 Stephen J. Brown, William N. Goetzmann, and Roger G. Ibbotson. "Offshore Hedge Funds: Survival and Performance, 1989–95". In: *Journal of Business* 72.1 (1999), pp. 91–117.

29 Markus K. Brunnermeier and Stefan Nagel. "Hedge Funds and the Technology Bubble". In: *Journal of Finance* 59.5 (Oct. 2004), pp. 2013–2040.

30 Charles Cao, Yong Chen, Bing Liang, and Andrew W. Lo. "Can Hedge Funds Time Market Liquidity?" In: *Journal of Financial Economics* 109.2 (2013), pp. 493–516.

31 Charles Cao, Bing Liang, Andrew W. Lo, and Lubomir Petrasek. "Hedge Fund Holdings and Stock Market Efficiency". In: *Review of Asset Pricing Studies* 8.1 (2018), pp. 77–116.

32 Mark M. Carhart. "On Persistence in Mutual Fund Performance". In: *Journal of Finance* 52.1 (Mar. 1997), pp. 57–82.

33 Gavin Cassar and Joseph Gerakos. "Hedge Funds: Pricing Controls and the Smoothing of Self-Reported Returns". In: *Review of Financial Studies* 24.5 (2011), pp. 1698–1734.

34 Ilia D. Dichev and Gwen Yu. "Higher Risk, Lower Returns: What Hedge Fund Investors Really Earn". In: *Journal of Financial Economics* 100.2 (2011), pp. 248–263.

35 William Fung and David A. Hsieh. "Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds". In: *Review of Financial Studies* 10.2 (Summer 1997), pp. 275–302.

36 William Fung and David A. Hsieh. "Performance Characteristics of Hedge Funds and Commodity Funds: Natural vs. Spurious Biases". In: *Journal of Financial and Quantitative Analysis* 35.2 (Apr. 2000), pp. 313–341.

37 William Fung and David A. Hsieh. "The Risk in Hedge Fund Strategies: Theory and Evidence from Trend Followers". In: *Review of Financial Studies* 14.2 (Summer 2001), pp. 313–341.

38 William Fung and David A. Hsieh. "Hedge Fund Benchmarks: A Risk-Based Approach". In: *Financial Analysts Journal* 60.5 (September–October 2004), pp. 65–80.

39 William Fung, David A. Hsieh, Narayan Y. Naik, and Tarun Ramadorai. "Hedge Funds: Performance, Risk, and Capital Formation". In: *Journal of Finance* 63.4 (2008), pp. 1777–1803.

40 Mila Getmansky. "The Life Cycle of Hedge Funds: Fund Flows, Size, Competition, and Performance". In: *Quarterly Journal of Finance* 2.1 (2012), p. 1250003.

41 Mila Getmansky, Peter A. Lee, and Andrew W. Lo. "Hedge Funds: A Dynamic Industry in Transition". In: *Annual Review of Financial Economics* 7 (2015), pp. 483–577.

42 Mila Getmansky, Andrew W. Lo, and Igor Makarov. "An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns". In: *Journal of Financial Economics* 74.3 (Dec. 2004), pp. 529–609.

43 Lawrence R. Glosten and Ravi Jagannathan. "A Contingent Claim Approach to Performance Evaluation". In: *Journal of Empirical Finance* 1.2 (1994), pp. 133–160.

44 William N. Goetzmann, Jonathan E. Ingersoll, and Stephen A. Ross. "High-Water Marks and Hedge Fund Management Contracts". In: *Journal of Finance* 58.4 (Aug. 2003), pp. 1685–1718.

45 M. Barry Goldman, Howard B. Sosin, and Mary Ann Gatto. "Path Dependent Options: "Buy at the Low, Sell at the High"". In: *Journal of Finance* 34.5 (1979), pp. 1111–1127.

46 John M. Griffin and Jin Xu. "How Smart Are the Smart Guys? A Unique View from Hedge Fund Stock Holdings". In: *Review of Financial Studies* 22.7 (May 2009), pp. 2531–2570.

47 Mark Grinblatt, Gergana Jostova, Lubomir Petrasek, and Alexander Philipov. "Style and Skill: Hedge Funds, Mutual Funds, and Momentum". In: *Management Science* 66.12 (Dec. 2020), pp. 5505–5531.

48 Mark Grinblatt, Sheridan Titman, and Russ Wermers. "Momentum Investment Strategies, Portfolio Performance, and Herding: A Study of Mutual Fund Behavior". In: *The American Economic Review* 85.5 (Dec. 1995), pp. 1088–1105.

49 Murali Jagannathan, Alexey Malakhov, and Dmitry Novikov. "Do Hot Hands Exist among Hedge Fund Managers? An Empirical Evaluation". In: *Journal of Finance* 65.1 (Feb. 2010), pp. 217–255.

50 Juha Joenväärä, Mikko Kauppila, Robert Kosowski, and Pekka Tolonen. "Hedge Fund Performance: Are Stylized Facts Sensitive to Which Database One Uses?" In: *Critical Finance Review* 10.2 (2021), pp. 271–327.

51 Philippe Jorion and Christopher Schwarz. "The Fix Is In: Properly Backing out Backfill Bias". In: *Review of Financial Studies* 32.12 (2019), pp. 5048–5099.

52 Petri Jylhä, Kalle Rinne, and Matti Suominen. "Do Hedge Funds Supply or Demand Liquidity?" In: *Review of Finance* 18.4 (July 2014), pp. 1259–1298.

53 Amir E. Khandani and Andrew W. Lo. "What Happened to the Quants in August 2007?" In: *Journal of Investment Management* 5.4 (2007), pp. 5–54.

54 Joni Kokkonen and Matti Suominen. "Hedge Funds and Stock Market Efficiency". In: *Management Science* 61.12 (2015), pp. 2890–2904.

55 Robert Kosowski, Narayan Y. Naik, and Melvyn Teo. "Do Hedge Funds Deliver Alpha? A Bayesian and Bootstrap Analysis". In: *Journal of Financial Economics* 84.1 (Apr. 2007), pp. 229–264.

56 Robert Kosowski, Allan Timmermann, Russ Wermers, and Hal White. "Can Mutual Fund "Stars" Really Pick Stocks? New Evidence from a Bootstrap Analysis". In: *Journal of Finance* 61.6 (2006), pp. 2551–2595.

57 Bing Liang. "Hedge Funds: The Living and theDead". In: *Journal of Financial and Quantitative Analysis* 35.3 (Sept. 2000), pp. 309–326.

58 Jongha Lim, Berk A. Sensoy, and Michael S. Weisbach. "Indirect Incentives of Hedge Fund Managers". In: *Journal of Finance* 71.2 (2016), pp. 871–918.

59 Andrew W. Lo. "The Statistics of Sharpe Ratios". In: *Financial Analysts Journal* 58.4 (2002), pp. 36–52.

60 Burton G. Malkiel and Atanu Saha. "Hedge Funds: Risk and Return". In: *Financial Analysts Journal* 61.6 (2005), pp. 80–88.

61 Robert C. Merton. "On market timing and investment performance. I. An equilibrium theory of value for market forecasts". In: *Journal of Business* 54 (1981), pp. 363–406.

62 Mark Mitchell, Lasse Heje Pedersen, and Todd Pulvino. "Slow Moving Capital". In: *American Economic Review* 97.2 (2007), pp. 215–220.

63 Mark Mitchell and Todd Pulvino. "Characteristics of Risk and Return in Risk Arbitrage". In: *The Journal of Finance* 56.6 (Dec. 2001), pp. 2135–2175.

64 Stavros Panageas and Mark M. Westerfield. "High-Water Marks: High Risk Appetites? Convex Compensation, Long Horizons, and Portfolio Choice". In: *Journal of Finance* 64.1 (2009), pp. 1–36.

65 Andrew J. Patton and Tarun Ramadorai. "On the High-Frequency Dynamics of Hedge Fund Risk Exposures". In: *Journal of Finance* 68.2 (2013), pp. 597–635.

66 Andrew J. Patton, Tarun Ramadorai, and Michael Streatfield. "Change you Can Believe In? Hedge Fund Data Revisions". In: *Journal of Finance* 70.3 (June 2015), pp. 963–999.

67 Ronnie Sadka. "Liquidity Risk and the Cross-Section of Hedge-Fund Returns". In: *Journal of Financial Economics* 98.1 (2010), pp. 54–71.

68 William F. Sharpe. "Asset Allocation: Management Style and Performance Measurement". In: *Journal of Portfolio Management* 18.2 (Jan. 1992), pp. 7–19.

69 Melvyn Teo. "The Liquidity Risk of Liquid Hedge Funds". In: *Journal of Financial Economics* 100.1 (2011), pp. 24–44.